

Human-Centered Design for AI

Target outcome:

Participants stop building solutions for themselves and start designing for humans.



What is HCD for AI?

- **Definition:** A framework that ensures AI solutions are grounded in human needs rather than just technical possibilities.
- **The Goal:** Moving from "What *can* we build with this data?" to "What *should* we build for this person?"
- **The HCD Loop:**
 - **Empathize:** Research user experiences.
 - **Define:** Identify the core problem.
 - **Ideate:** Brainstorm AI and non-AI solutions.
 - **Prototype:** Build low-fidelity versions.
 - **Test:** Get feedback and iterate.

Needs vs. Capabilities

- **Technical Capability:** Can we predict churn with 95% accuracy? (The "Math" side).
- **User Need:** Does the account manager know *why* the customer is leaving and *how* to talk to them? (The "Human" side).
- **The Sweet Spot:** Innovation happens at the intersection of **Desirability** (User), **Feasibility** (Tech), and **Viability** (Business).

User Research Fundamentals

- **Interviews:** Ask "Why?" to uncover underlying motivations. Avoid leading questions.
- **Observations (Contextual Inquiry):** Watch how users currently solve the problem. Often, they have "hacks" that reveal more than their words do.
- **Key Tip:** Look for "Extreme Users"—those who use your tool in ways you didn't intend.



Personas & Journey Maps

- **User Personas:** Not just a bio, but a set of **behaviours, goals, and pain points**.
 - *Example:* "Data-Overwhelmed David" needs summaries, not raw tables.
- **Journey Mapping:** Visualising the process a user goes through to accomplish a goal.
 - Identify the "**Moan Moments**" (Pain points) where AI can provide the most relief.

Ethical Design & Responsibility

- **Accessibility:** Is the AI output readable for those with visual impairments?
- **Inclusivity:** Does the training data represent all user demographics fairly?
- **Cultural Sensitivity:** Does the AI's tone or recommendation make sense in a global context?
- **Transparency:** Can the user understand *why* the AI made a specific suggestion? (Explainable AI).



Workshop: Rapid HCD Lab

Activity 1: Rapid Research (10 Mins)

- **Task:** Pair up. One person is the "User," the other is the "Researcher."
- **Goal:** Interview your partner about the problem your AI project aims to solve.
- **Ask:** "Tell me about the last time you dealt with [Problem]. What was the hardest part? Why?"



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Activity 2: Persona & Journey Map (15 Mins)

- **Create a Persona:** Give them a name, a primary goal, and their biggest frustration with current tech.
- **Map the Journey:** List the 4–5 steps they take now.
- **Identify Pain Points:** Use a red pen to circle where they get stuck or frustrated.

Activity 3: AI Opportunity Identification (10 Mins)

- **Task:** Brainstorm 3 AI features that directly address the red circles on your map.
 - *Example:* Instead of "A Dashboard," maybe the solution is "An AI alert that triggers when [X] happens."
- **Filter:** Choose the one that balances user impact with your current technical data.



Workshop: Rapid HCD Lab



Sharing & Feedback

- **The Pitch:**
 - Present your persona and their journey to another pair.
- **Constraint:** You have 60 seconds to explain:
 - Who is this for?
 - What is their pain? and
 - How does your AI heal it?

